YAKEEN

NEET 2.0 2027

Physics | Lecture - 08

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Todays Goal

Variation of slope
& Quadratic eqn.

differentiation ✓

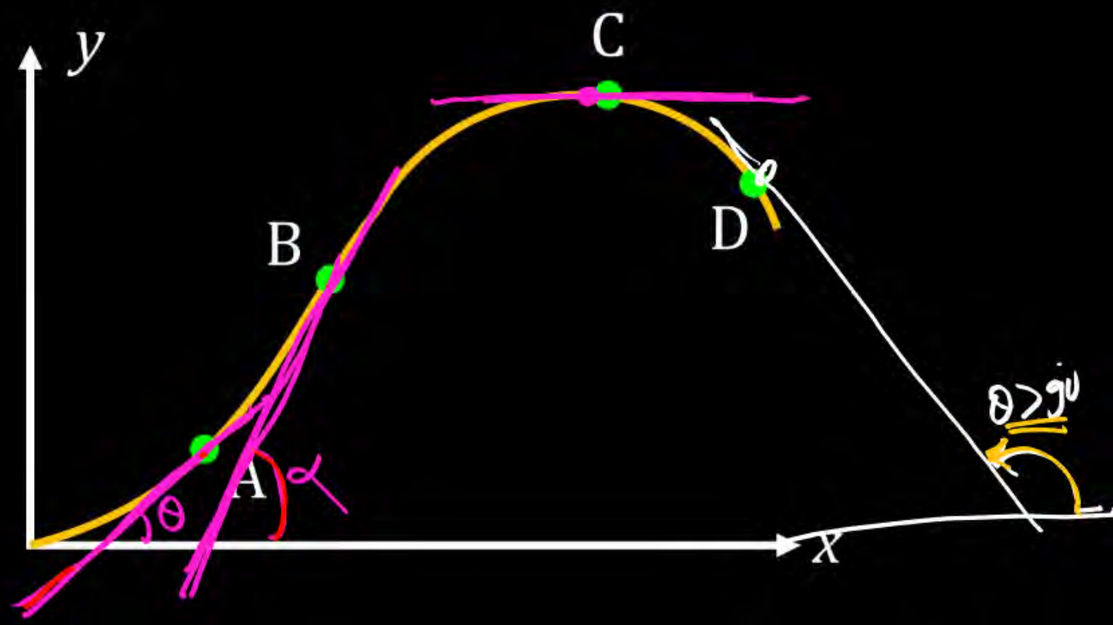
Sangha assignment

→ 3rd graph. Quad / A|P | G.P

Question

Match the matrix:

Point	Slope
A	Zero
B	negative
C	Maximum
D	Positive



Question

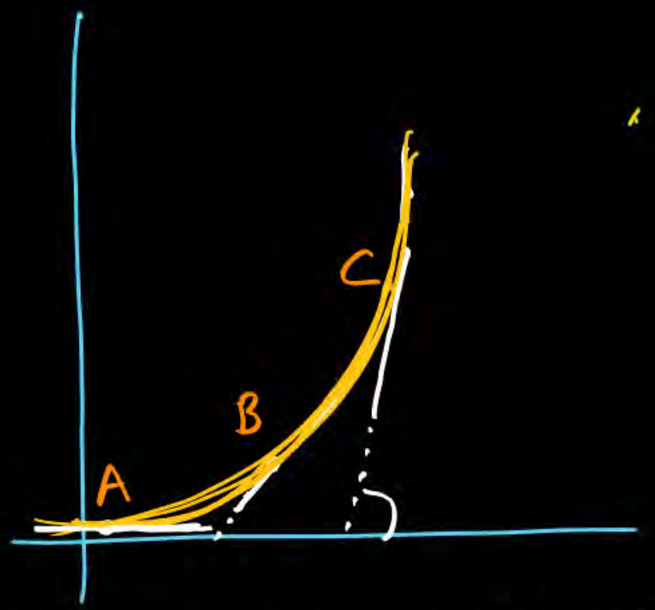
If $y = 2x + 4$ find change in y w.r.t x, when x changes from $x_1 = 1$ to $x_2 = 3$



Draw tangent at a point where slope have to find.
then slope = $\tan\theta$ where θ = Angle of tangent with x-axis.

slope at any point $\boxed{\tan\theta = \frac{dy}{dx}}$; diff^{n} of y w.r.t. x.

(2)

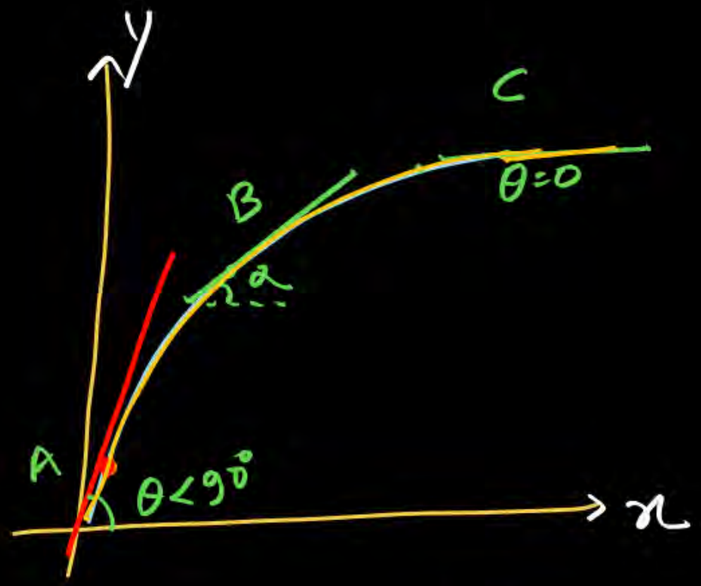


$m_A = 0$ m_B = +ve m_C = +ve

\# $m_A < m_B < m_C$

① Slope = +ve and ↑

② Magnitude of slope (↑)



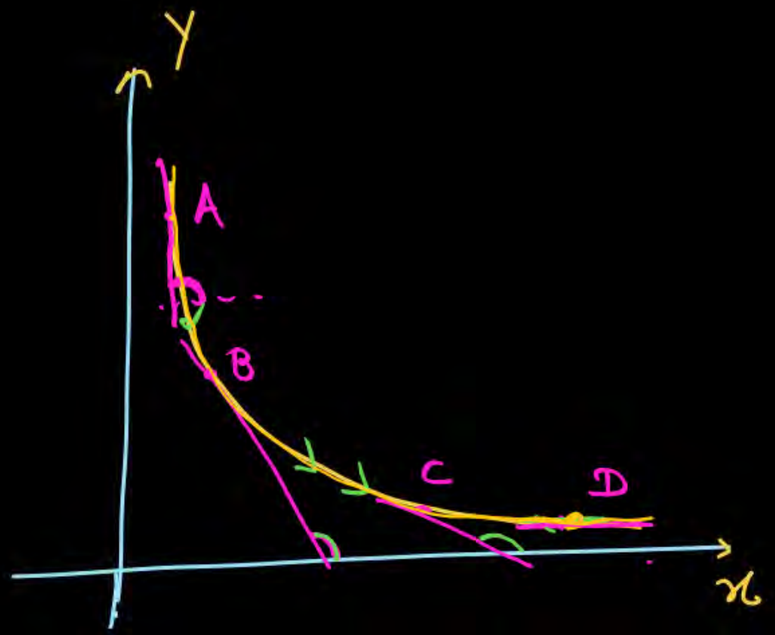
m_A = +ve $m_C = 0$

m_B = +ve

$m_A > m_B > m_C$

① Slope is +ve & ↓

② Magnitude of slope (↓)

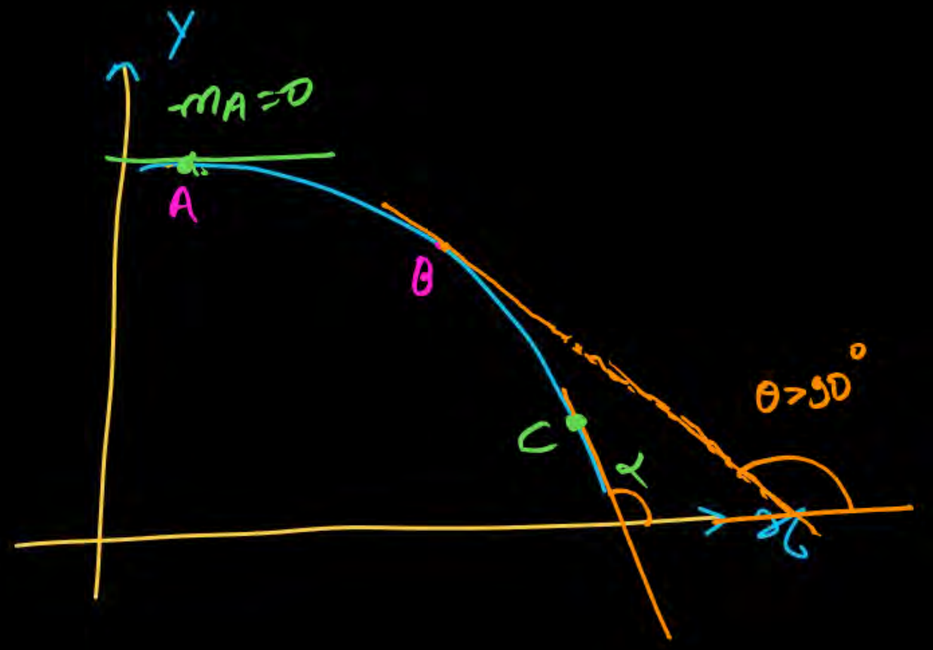


$m_D = 0$

$$\begin{cases} m_A = -ve \\ m_B = -ve \\ m_C = -ve \end{cases}$$

⊙ Slope is ↑ (Increasing)

• Magnitude of slope (↓)



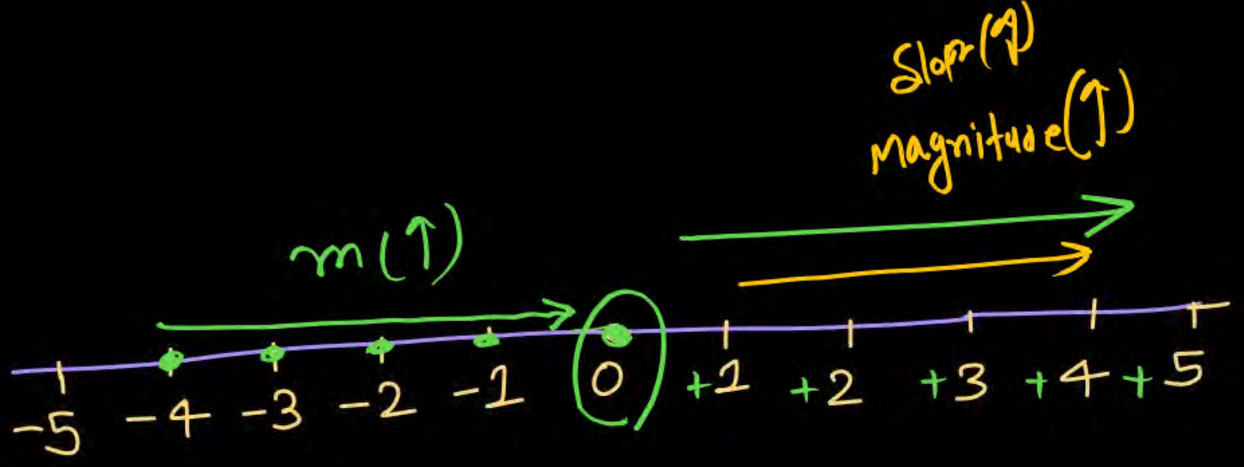
$m_A = 0$ $m_B = -ve$

$m_C = -ve$

① Slope $(0, -2, -2, -3)$ ↓

magnitude of slope (↑)

Variation of slope and magnitude of slope :-



Magnitude of slope (↓)

-4 -3 -2 -1 0

→ Increasing

MR* Box

for slope



हसता हुआ
"Ramlal"



रोता हुआ
"Ramlal"

Slope ↑

Slope ↓

MR* Box for

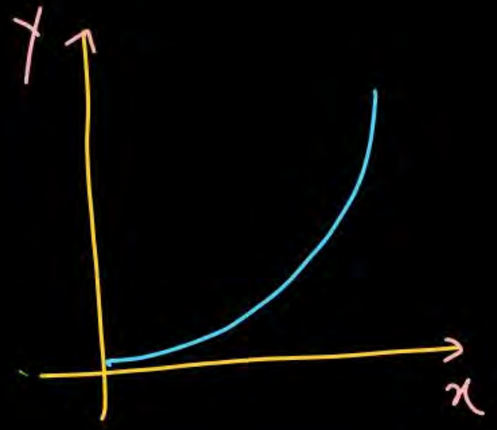
Magnitude of Slope

Horizontal & Vertical line

dekho → Horizontal Ki

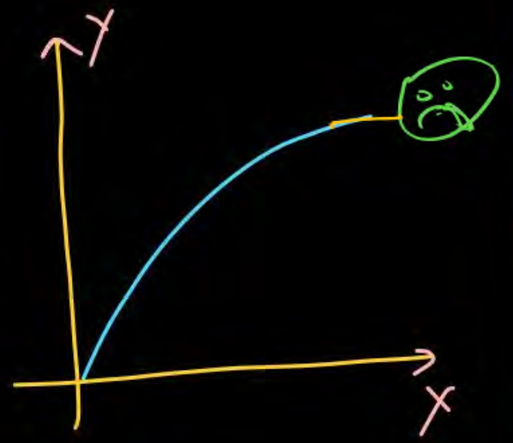
Traf Jane se |m| ↓ decreasing

Horizontal se dur |m| ↑ increasing



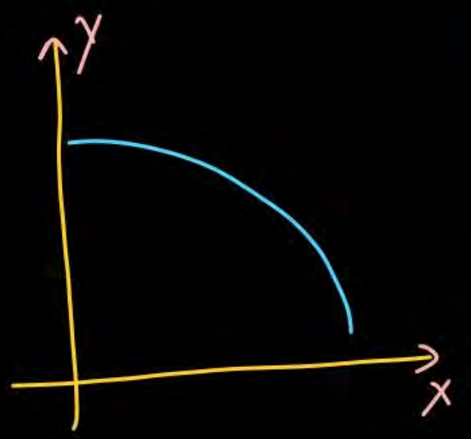
slope = ↑

magnitude of slope = ↑



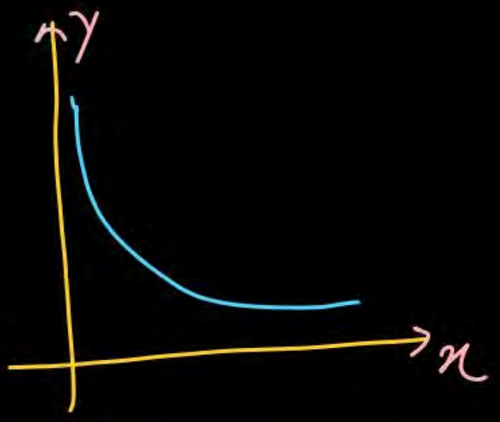
slope = ↓

magnitude of slope = ↓



slope = ↓

magnitude of slope = ↑



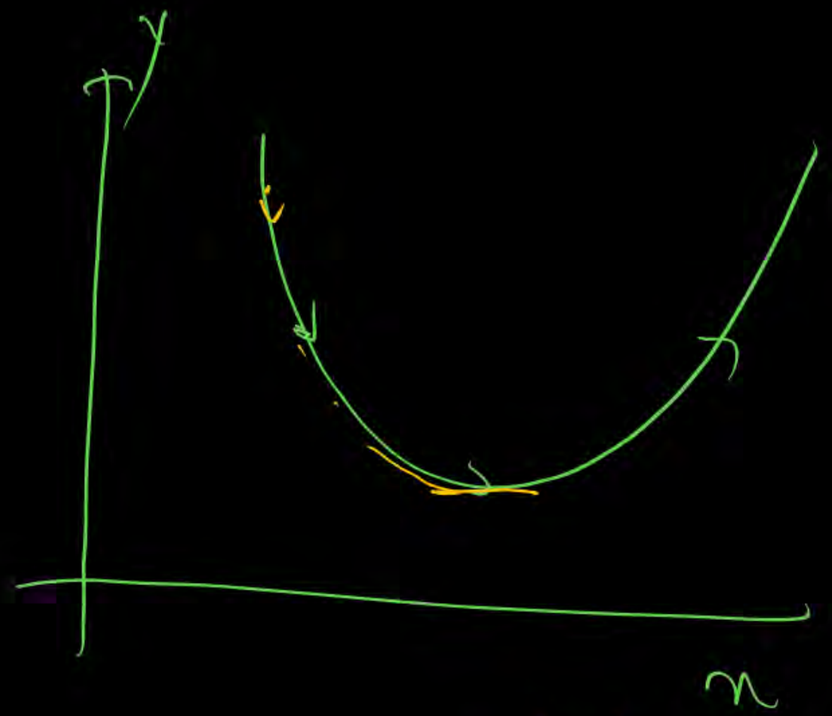
slope = ↑

magnitude of slope = ↓



Slope is continously ↓

⊙ magnitude of slope 1st ↓ (dec) then ↑ (Inc)



Slope (↑)

magnitud of slope 1st (↓) then (↑)

Quadratic equation

$$ax^2+bx+c=0$$

$\Rightarrow$ a can't be zero for quadratic eq.

* $x_1 = \dfrac{-b+\sqrt{b^2-4ac}}{2a}$

* $x_2 = \dfrac{-b-\sqrt{b^2-4ac}}{2a}$

⊙ If $b^2 = 4ac$, then only one root is possible $x_1 = x_2 = -\dfrac{b}{2a}$

⊙ If $b^2 > 4ac$ then there is Two root x_1 & x_2

⊙ If $b^2 < 4ac$ then there is no real root

⊙ Sum of root

$$x_1 + x_2 = -\frac{b}{2a} - \frac{b}{2a} = -\frac{b}{a}$$

⊙ Product of roots $= \dfrac{c}{a}$

$x^2 - 7x + 12 = 0$ find roots of equation

$ax^2 + bx + c = 0$

$a = 1 \quad b = -7 \quad c = 12$

method-1

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{+7 + \sqrt{49 - 4 \times 1 \times (12)}}{2 \times 1}$$

$$\begin{cases} x_1 = \dfrac{7 + \sqrt{1}}{2} = \dfrac{7+1}{2} = 4 \\ x_2 = \dfrac{7 - \sqrt{1}}{2} = \dfrac{6}{2} = 3 \end{cases}$$

$$x_1 + x_2 = \frac{-b}{a} = \frac{-(-7)}{1} = 7$$

method-2

$x^2 - 7x + 12 = 0$

$x^2 - 3x - 4x + 12 = 0$

$x(x-3) - 4(x-3) = 0$

$(x-3)(x-4) = 0$

$x - 3 = 0$ | $x - 4 = 0$

$x = 3$ | $x = 4$ ✓

Q) $x^2 - 7x - 18 = 0$

find roots of equation

$x^2 - 9x + 2x - 18 = 0$

$x(x-9) + 2(x-9) = 0$

$(x-9)(x+2) = 0$

$x_1 = 9 \quad x_2 = -2$

(a) $x^2 + 7x - 18 = 0$

$x^2 + 9x - 2x - 18 = 0$

$x(x+9) - 2(x+9) = 0$

$x_1 = -9$

$x_2 = +2$

(a) $x^2 + 8x + 15 = 0$

Soln $x^2 + 5x + 3x + 15 = 0$

$x(x+5) + 3(x+5) = 0$

$(x+5)(x+3) = 0$

$x+5 = 0 \quad x+3 = 0$

$x_1 = -5 \quad x_2 = -3$

(Q) $y = x^2 - 3x - 10$ Draw graph b/w y & x

→ Parabola

Put $y = 0$

$0 = x^2 - 3x - 10$

$x^2 - 5x + 2x - 10 = 0$

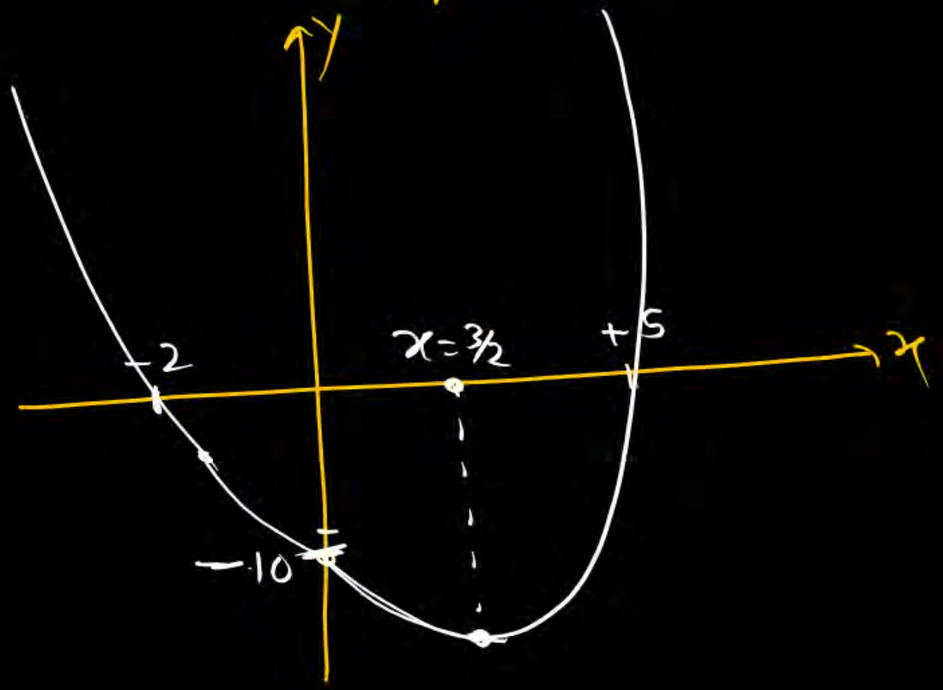
$x(x-5) + 2(x-5) = 0$

$(x-5)(x+2) = 0$

$x_1 = 5$ $x_2 = -2$

Put $x = 0$

$y = -10$



" gf y will be max^m or min^m when $\frac{dy}{dx} = 0$ "

$y = x^2 - 3x - 10$

$\frac{dy}{dx} = 2x - 3 = 0$

$2x - 3 = 0$

$x_1 = \frac{3}{2}$

(11) $y = +x^2 - 9x + 14$ Draw graph b/w y & x

y intercept ($x=0$) = $+14$

x → Intercept ($y=0$)

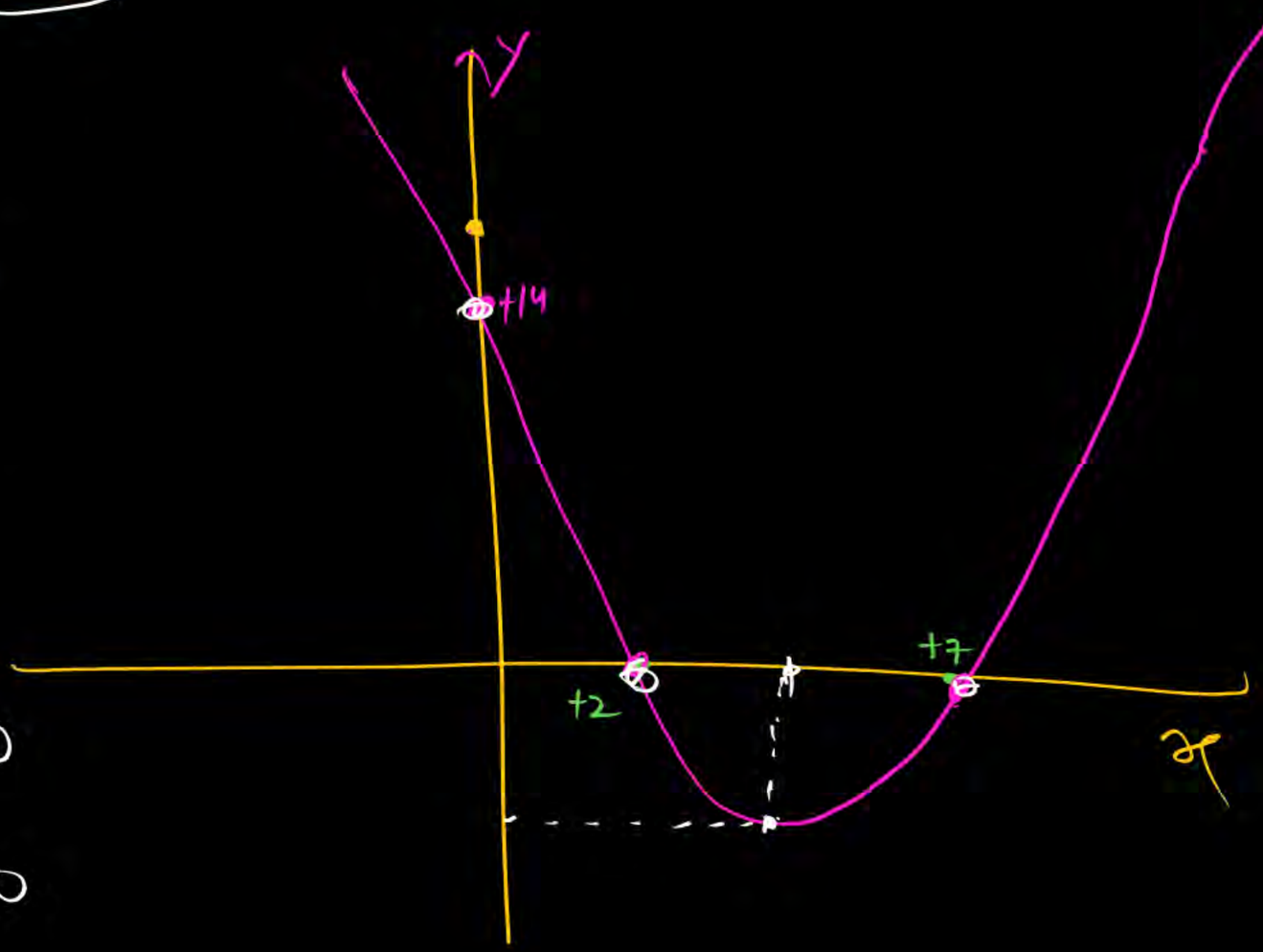
$$0 = x^2 - 9x + 14$$

$$x^2 - 7x - 2x + 14 = 0$$

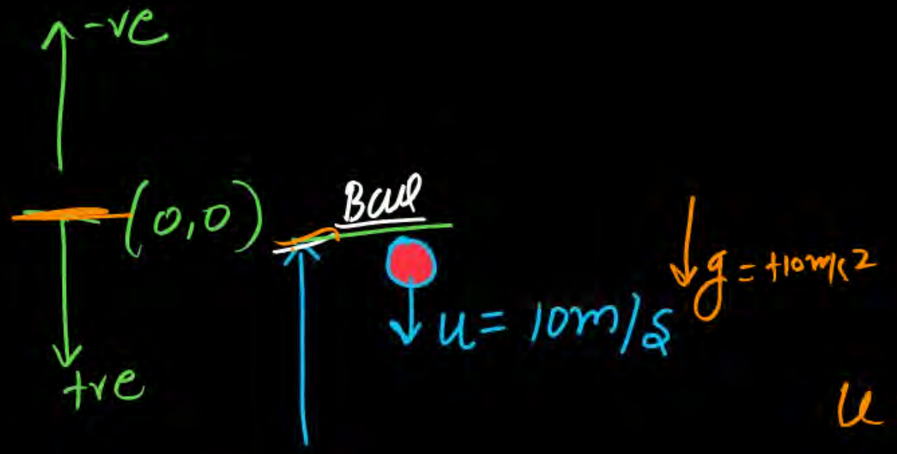
$$x(x-7) - 2(x-7) = 0$$

$$(x-7)(x-2)$$

$x_1 = +7 \quad x_2 = +2$



(Q) Ball is thrown downward with velocity 10 m/s from a tower of Height 75m then find time of flight.



$u = +10\ m/s$

$a = +10\ m/s^2$

$s = +75\ m$

$s = ut + \frac{1}{2}at^2$

$75 = 10t + \frac{1}{2} \times 10\, t^2$

$5t^2 + 10t - 75 = 0$

$t^2 + 2t - 15 = 0$

$t^2 + 5t - 3t - 15 = 0$

$t(t+5) - 3(t+5) = 0$

$(t+5)(t-3) = 0$

$t_1 = 3$ $\quad t_2 = -5$

$t = +3$ sec Ans

$t = \frac{\text{distn}}{\text{speed}}$

$= \frac{75}{10}$

$t = 7.5$

If $y = 3x - 2$ then find

(i) value of y at $(x = 3)$ $\quad y = 3 \times 3 - 2 = 7$

(ii) value of y at $(x = 8)$ $\quad y = 3 \times 8 - 2 = 22$

(iii) find change in y when x change from $x_1 = 1$ to $x_2 = 3$

$$y_1(x_1 = 1) = 3 \times 1 - 2 = 1 \qquad y_2(x_2 = 3) = 7$$

$$\text{Change in } y = y_2 - y_1 = 7 - 1 = 6$$

(iv) find change in y w.r.t change in x when $x_1 = 1$ $x_2 = 3$

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7 - 1}{3 - 1} = \frac{6}{2} = 3$$

(v) find change in y when x change from $x_1 = 2$ to $x_2 = 2.00001$

Soln

$$\frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

differentiation

$\Delta x \to$ very small

$\frac{d}{dx}$ = differential operation

$\frac{d\,\square}{dx}$ = The rate of change in $\square$ w.r.t. x

$\Rightarrow \frac{d\,[Y]}{dx}$ = The rate of change in $[Y]$ w.r.t. x

$\Rightarrow \frac{d\,[P]}{dx}$ = The rate of change in $[P]$ w.r.t. x

$\Rightarrow \frac{d\,[X]}{dt}$ = The rate of change in $[X]$ w.r.t. 't'

$\frac{dy}{dx}$ ✗

$\frac{d(y)}{dx}$ ✓

(Q) If $y = 2x + 4$ then find

(1) Value of y at $x = 1$

(2) Value of y at $x = 3$

$\frac{d}{dx}$ = differential operator

$\frac{d\square}{dx}$ = The rate of change in $\square$ w.r.t x

$\frac{d\square}{dx}$ = The rate of change in $\square$ w.r.t x

$\frac{d\square}{dx}$ = The rate of change in $\square$ w.r.t x

Rule of differentiation:

Rule 1. Differentiation of a constant function is zero

$$\frac{dy}{dx} = 0 \quad ; \quad y = \text{cost}^n$$

$$\Rightarrow \frac{d\pi}{dx} = 0$$

$$\Rightarrow \frac{dG}{dt} = 0$$

← gravitational fixed

$$\frac{d(5)}{dt} = 0$$

physics



$E_{inside} = 0$

$$\vec{E} = -\frac{dV}{dr} = 0$$

$$\frac{d(\sin^2\theta + \cos^2\theta)}{d\theta} = \frac{d1}{d\theta} = 0$$

$$\frac{d(\tan\theta \cdot \cot\theta)}{d\theta} = 0$$

$$\frac{d\,2}{dt} = 0 \quad \text{(Constant)}$$

$$\frac{d(x^0)}{dx} = 0$$

$$\frac{d(\sin 45^\circ)}{d\theta} = \cos(45^\circ)$$

$$\frac{d(1/\sqrt{2})}{d\theta} = 0 \checkmark$$

Rule - 2 Differentiation of algebric function

$$y = x^n$$

$$\frac{d(y)}{dx} = \frac{d(x^n)}{dx} = n x^{(n-1)}$$

Ex

$$\frac{d(x^2)}{dx} = 2x^{2-1} = 2x^1$$

$$\frac{d(x^5)}{dx} = 5x^4$$

$$\frac{dx^{3/2}}{dx} = \frac{3}{2} x^{\frac{3}{2}-1}$$

$$\frac{d(x^{1/3})}{dx} = \frac{1}{3} x^{\frac{1}{3}-1}$$

$$\frac{d\sqrt{x}}{dx} = \frac{dx^{1/2}}{dx} = \frac{1}{2} x^{\frac{1}{2}-1}$$

$$= \frac{1}{2} x^{-\frac{1}{2}}$$

AS

Rule-3 Diff^n of exponential / Trigo / log.

$$\frac{d(e^x)}{dx} = e^x$$

$$\frac{d(\sin\theta)}{dx}$$

$$\frac{d\sin(\theta)}{d\theta} = \cos\theta$$

$$\frac{d(\sec\theta)}{d\theta} = \sec\theta \cdot \tan\theta$$

$$\frac{d(\cos\theta)}{d\theta} = -\sin\theta$$

$$\frac{d\log e^x}{dx} = \frac{1}{x}$$

$$\frac{d(\tan\theta)}{d\theta} = \sec^2\theta$$

Rule - 4 If any Constant multiply with Variable then Constant comes out from differentiation.

Ex $y = 3x^5$

(or)

$$\frac{dy}{dx} = \frac{d(3x^5)}{dx}$$

$$= 3\frac{dx^5}{dx}$$

$$= 3 \times 5x^{5-1}$$

$$= 15x^4$$

Rule → 5 addition Rule

Ex- $y = A + B$

$$\frac{dy}{dx} = \frac{dA}{dx} + \frac{dB}{dx}$$

Ex $y = \sin x + x^2$

$$\frac{dy}{dx} = \frac{d(\sin x)}{dx} + \frac{d(x^2)}{dx}$$

$$= \cos x + 2x$$

Rule-6 Subtraction Rule

$$y = A - B$$

$$\frac{dy}{dx} = \frac{dA}{dx} - \frac{dB}{dx}$$

$$y = e^x - \sin x$$

$$\frac{dy}{dx} = \frac{de^x}{dx} - \frac{d(\sin x)}{dx}$$

$$= e^x - \cos x$$

Rule - 7 Product Rule

$$y = A \cdot B$$

$$\frac{dy}{dx} = \frac{dA}{dx} B + B \frac{dA}{dx}$$

Rule 8 division Rule

$$y = \frac{A}{B}$$

$$\frac{dy}{dx} = \frac{B \frac{dA}{dx} - A \frac{dB}{dx}}{B^2}$$

H/W

Copy in Notes

$$y = \tan x \cdot \log x$$

(Q) $y = \frac{A}{x^3} - \frac{B}{x^4}$ find x where $\frac{dy}{dx} = 0$

$$y = \frac{e^x}{x^2}$$

$$y = \frac{x^2 - 1}{x^2 + 1}$$

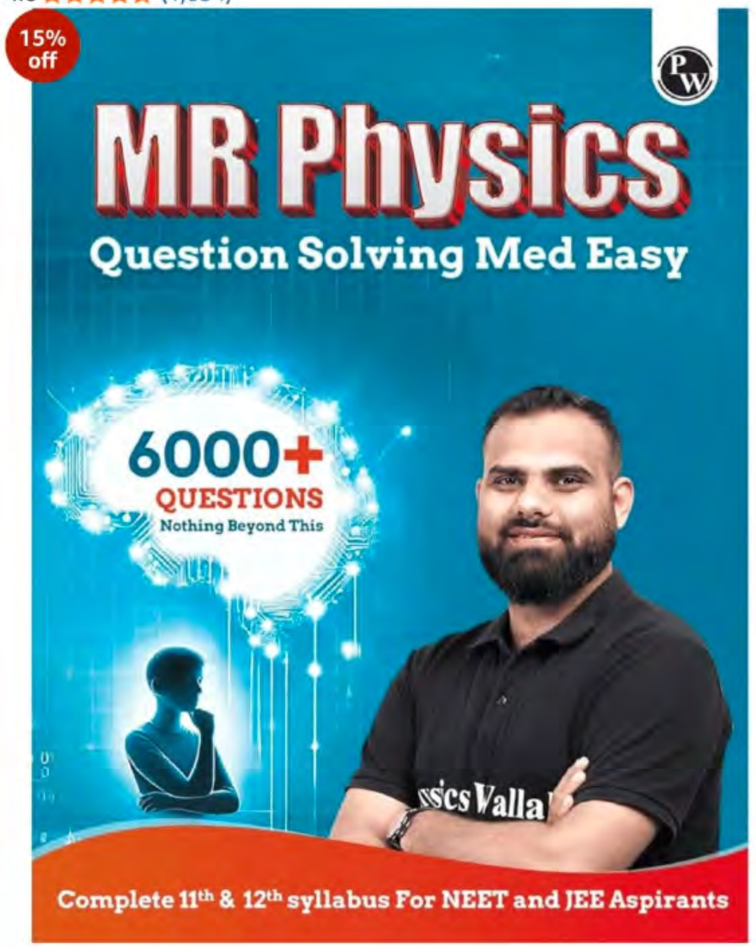
$$y = \cos x + \frac{2}{x^3}$$

$$y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2$$

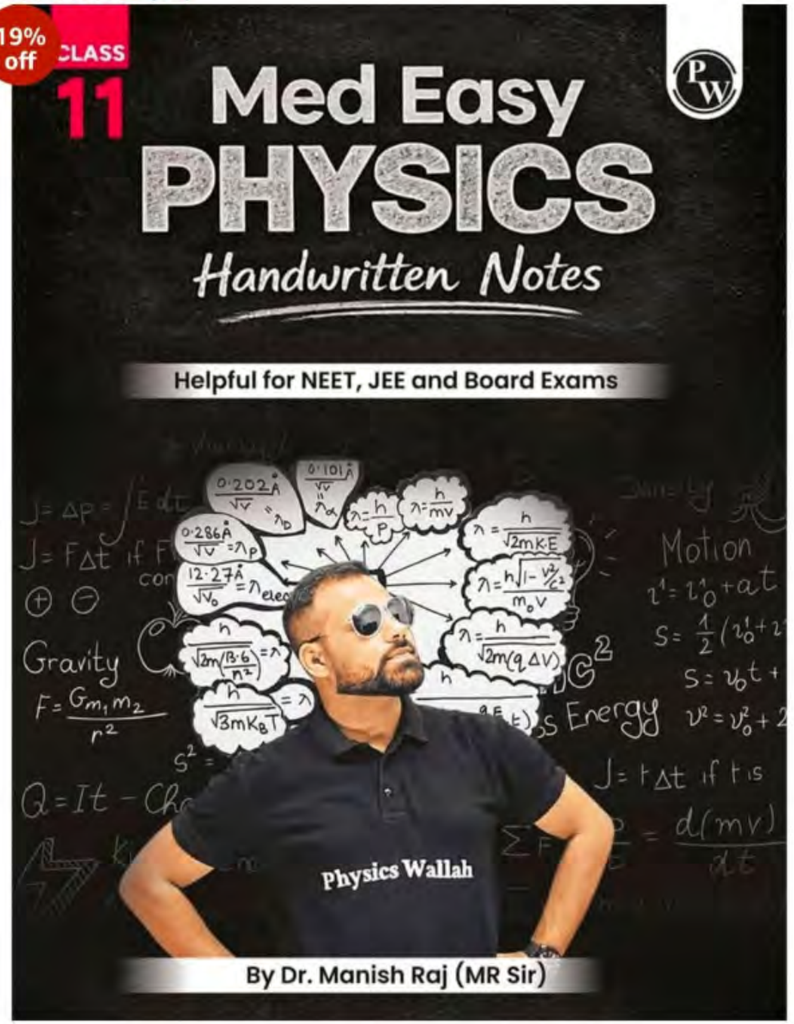
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4.6 ★★★★½ (1,054)



4.6 ★★★★½ (13)



Thank You